

PROBLEM A: Gerrymandering The United States Constitution provides that the House of Representatives shall be composed of some number (currently 435) of individuals who are elected from each state in proportion to the state's population relative to that of the country as a whole. While this provides a way of determining how many representatives each state will have, it says nothing about how the district represented by a particular representative shall be determined geographically. This oversight has led to egregious (at least some people think so, usually not the incumbent) district shapes that look "unnatural" by some standards.

Hence the following question: Suppose you were given the opportunity to draw congressional districts for a state. How would you do so as a purely "baseline" exercise to create the "simplest" shapes for all the districts in a state? The rules include only that each district in the state must contain the same population. The definition of "simple" is up to you; but you need to make a convincing argument to voters in the state that your solution is fair. As an application of your method, draw geographically simple congressional districts for the state of New York.

A 题：不公平的选区划分

美国宪法规定众议院由一定数目的众议员(目前是 435 人)组成,他们是由各州按照该州人口占全国总人口的百分比选出来的。尽管这种规定提供了确定每个州有多少众议员的方法,但是一点也没有说及有关一个特定的众议员所代表的选区应该怎样按地区决定的问题。这种疏忽已经导致了按某种标准看来是违反常情的很不好的(至少某些人认为通常是不必这样做的)选区安排。

因此就向你们提出了以下的问题:假设你们有机会去制定一个州的众议院的选区。你们会怎样把它作为一种纯"基础性"的练习来创建一个州的所有选区的"最简单"的划分。这些划分规则中至少要包含一条:该州的每个选区必须有同样的人口。"简单"的定义要由你们来下;但是你们必须就你们的解决方法是公正的做出一个能够使该州选民信服的论证。作为你们的方法的应用,试创建纽约州的按地域来说是简单的选区划分。